

Unit III TREE STRUCTURES

Course objective: To understand the concepts of tree

Course Outcome: Implement in expression tree, binary search, AVL, B tree, and priority queue

Bloom's Taxonomy Level: K1, K2 and K3

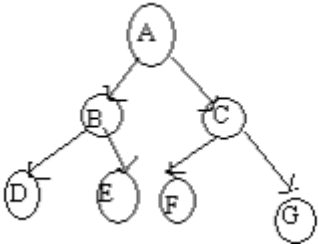
PART A

Q.N	Question	Bloom's Taxonomy Level
1	Define non-linear data structure Data structure which is capable of expressing more complex relationship than that of physical adjacency is called non-linear data structure.	K1
2	Define tree Data Structures? • A tree is non-linear and a hierarchical data structure consisting of a collection of nodes. • Specialized method to organize and store data in the computer to be used more effectively. • It consists of a central node, structural nodes, and sub-nodes, which are connected via edges. We can also say that tree data structure has roots, branches, and leaves connected with one another.	K1
3	List out the steps involved in deleting a node from a binary search tree. 1. t has no right hand child node $t \rightarrow r == z$ 2. t has a right hand child but its right hand child node has no left sub tree $t \rightarrow r \rightarrow l == z$ 3. t has a right hand child node and the right hand child node has a left hand child node $t \rightarrow r \rightarrow l != z$	K1
4	Define Ancestor of a Node. Any predecessor nodes on the path of the root to that node are called Ancestors of that node.	K1
5	Convert the infix expression $(A-B/C)*(D/E-F)$ into a postfix. Postfix: $ABC/-DE/F-*$	K1
6	Define a Node. • A node is an entity that contains a key or value and pointers to its child nodes. • The last nodes of each path are called leaf nodes or external nodes that do not contain a link/pointer to child nodes.	K1

	•The node having at least a child node is called an internal node.	
7	What are the applications of binary tree? 1. Binary tree is used in data processing. 2. File index schemes 3. Hierarchical database management system	K1
8	What is meant by traversing? And its types. Traversing a tree means processing it in such a way, that each node is visited only once. The different types of traversing are a. Pre-order traversal-yields prefix form of expression. b. In-order traversal-yields infix form of expression. c. Post-order traversal-yields postfix form of expression.	K1

9	What is a balance factor in AVL trees? Balance factor of a node is defined to be the difference between the height of the node's left subtree and the height of the node's right subtree.	K1
10	Define pivot node. The node to be inserted travel down the appropriate branch track along the way of the deepest level node on the branch that has a balance factor of +1 or -1 is called pivot node.	K1
11	Define B tree A B-tree is a self-balancing tree data structure that maintains sorted data and allows searches, sequential access, insertions, and deletions in logarithmic time. The B-tree generalizes the binary search tree, allowing for nodes with more than two children.	K1

12	Define Full binary tree. A full binary tree is a binary tree in which all nodes except leaf nodes have two children. A full Binary tree is a special type of binary tree in which every parent node/internal node has either two or no children. It is also known as a proper binary tree.	K1
13	List out the advantages of B tree. 1. Maintain balanced height, ensuring efficient search, insertion and deletion operations with an average time complexity of $O(\log n)$. 2. It can store large amount of data, making them suitable for disk storage and database indexing. 3. B-Tree also support range queries and concurrent access, making them a popular choice for database indexing and storage.	K1

14	Traverse the given tree using Inorder, Preorder and Postorder traversals.  <pre> graph TD A((A)) --> B((B)) A --> C((C)) B --> D((D)) B --> E((E)) C --> F((F)) C --> G((G)) </pre> Inorder : D H B E A F C I G J Preorder: A B D H E C F G I J Postorder: H D E B F I J G C A	K2
15	Define a binary tree A binary tree is a finite set of nodes which is either empty or consists of a root and two disjoint binary trees called the left sub-tree and right sub-tree.	K1
16	What are the tasks performed during inorder traversal? • Traverse the left sub-tree • Process the root node • Traverse the right sub-tree	K1
17	State the merits of linear representation of binary trees. • Storage method is easy and can be easily implemented in arrays • When the location of a parent/child node is known, other one can be determined easily • It requires static memory allocation so it is easily implemented in all programming language	K1
18	Define Expression Tree. The expression tree is a tree used to represent the various expressions. The tree data structure is used to represent the expressional statements. In this tree, the internal node always denotes the operators.	K1
19	What is priority queue? A priority queue is a type of data structure that allows elements to be inserted and removed based on their priority. It's a queue where elements are ordered according to their priority and the highest priority element is served first.	
20	What is binary heap? A Binary Heap is either Min Heap or Max Heap. In a Min Binary Heap, the key at the root must be minimum among all keys present in Binary Heap. The same property must be recursively true for all nodes in Binary Tree. Max Binary Heap is similar to MinHeap.	K1

PART B

Q.NO	Question	Bloom's Taxonomy Level
1	Explain the AVL tree insertion and deletion with suitable example. (16 marks)	K2
2	Describe the algorithms used to perform single and double rotation on AVL tree. (16 marks)	K1
3	Explain about B-Tree with suitable example. (16 marks)	K2
4	Construct a B-tree of order 3 with each node can have at most 3 children from the following set of keys: {10, 20, 30, 40, 50, 60, 70, 80, 90} (16 marks)	K2
5	Explain the tree traversal techniques with an example. (16 marks)	K2
6	Construct an expression tree for the expression $(p+q*r) + ((s*t+u)*v)$. Give the outputs when you apply inorder, preorder and postorder traversals. (16 marks)	K3
7	(i) How to insert and delete an element into a binary search tree and write down the code for the insertion routine with an example. (8 marks) (ii) Define Min heap and max heap with an example. (8 marks)	K1
8	Construct a binary search tree for the following numbers start from an empty binary search tree. 55, 36, 20, 70, 80, 40, 50 Delete keys 20, 70 and 55 one after the other and show the trees at each stage. (16 marks)	K3